

Think-a-Dot Codea sketch written in Lua and run in the Codea IDE app.

Codea uses the [Processing.org](https://processing.org) sketch template:

- a `setup()` method that runs once and
- a `draw()` method that loops.

Codea supports iPad touch input and includes an extensive library of graphical routines for game development.

To run the sketch, install Codea on your iPad, iPhone, or MacBook (if it uses Apple's M-Series Chips), then press the play button. I've only tested it on my iPad, iPhone 12, and MacBook Air with the M2 CPU chip, and it works well.

Some further notes:

- The code was generated by ChatGPT 5.2 using vibe coding methods, where you describe to ChatGPT what the code should do and what it should look like, and hope for the best.
- Sometimes one or more runtime errors occur, I would take the error and ask ChatGPT to find and correct the mistake. But first, you would need to upload the sketch to ChatGPT and then ask it to fix the error.
- Once uploaded, you only need to provide new errors as they pop up
- Please always ask ChatGPT to generate the whole sketch, don't try to do the replace suggestions if you provide them. This minimizes mistakes on your part where you pasted the code in the wrong place.
- Codea can export code as an XCode project to make it run natively on Apple platforms

Gameplay

There are three white circular buttons at the top to drop a marble into one of three columns. The marble path is shown as it moves to the bottom—blue routes to the right and Yellow routes to the left. The bottom nodes indicate which side the marble will exit.

Tapping the left or right button is equivalent to tipping the Think-a-Dot to the left or right and sets the flipflops to either the yellow diamond on blue pattern or the blue diamond on yellow.

Tapping on the flipflop flips its color.

Acknowledgements:

The **Think-a-Dot** toy was designed by Joseph A. Weisbecker, an engineer at RCA (he worked on games in his spare time), and was later patented in the mid-1960s. It was then manufactured and sold by the educational toy company E.S.R., Inc. during the 1960s.

Think-a-Dot is covered by a U.S. patent: **US 3,388,483 A — titled Computer-type game and teaching aid** — filed on July 21, 1966, and granted on June 18, 1968, to Joseph A. Weisbecker. (see [think-a-dot-patent-US3388483-A.pdf](#))

John Godfrey, in the 1990s, provided an unofficial Think-a-Dot analysis and theory of operation manual (see [Think-a-Dot-Theory-by-John-Godfrey.pdf](#))

This sketch was generated by ChatGPT 5.2 using my Vibe coding methods on 12/23/2025.

Jim McArdle, 12/23/2025